Poster: GEM-PLI Side Channel in GPON: Inferring Encrypted Streaming Traffic

Hyunjun An
Heungdeok High School
hjahn@anhyunjun.com

***Abstract*—Gigabit Passive Optical Networks (GPON), an ITU-T standardized technology widely adopted in fiber-to-the-home (FTTH) networks, supports downstream payload encryption to protect subscribers' traffic. Unfortunately, encryption does not cover all access-layer metadata. This poster presents a preliminary analysis showing that the Payload Length Indicator (PLI), which remains in plaintext in GEM headers, forms a side channel that enables inference of streaming activity. Using a GPON downstream simulator and controlled runs of youtube streaming, our preliminary results show that PLI traces are highly consistent across repeated runs of the same video and suggest that a passive attacker on the same optical splitter as the victim could potentially infer a victim's streaming activity (e.g., Youtube Video Identification).**

I. Introduction And Motivation

As fiber internet access has become dominant in Network Infrastructures, GPON–one of the passive optical network (PON) technologies–is adopted by many ISPs to provide Fiber Internet for home (FTTH). Since GPON is popular PON technology, it is important to ensure traffic confidentiality.

GPON supports AES-based downstream traffic encryption on GPON Encapsulation Method (GEM) Layer. However, some fields of GEM Header remain plaintext such as the Payload Length Indicator (PLI) field used for multiplexing and the Port-ID field used for mapping subscribers' Optical Network Terminals/Units(ONTs/ONUs).

Such plaintext fields may appear innocuous. However, recent studies have demonstrated that such metadata can be exploited to infer victim's activity (e.g., identifying a victim's web-based video streaming watching activity) [4] [5] [6].

While prior studies primarily focused on upper-layer or cellular traffic analysis [4], [5], We researched whether the plaintext PLI field in GPON GEM headers can be used to leak victim's privacy on the same optical splitter. We conduct a preliminary experimental analysis using a software-based GPON downstream simulator to evaluate whether the plaintext PLI field could contribute to privacy leakage. Using these PLI traces, we present preliminary evidence that YouTube streaming traces are highly consistent across repeated runs and may be distinguishable in controlled settings.

II. Background

GPON Topology and GTC Framing GPON Networks consist of an Optical Line Terminal (OLT) at the ISP and multiple Optical Network Terminals/Units(ONTs/ONUs) (typically called "modem") deployed at subscribers. A single optical tree can serve up to 32 subscribers (e.g., 1:32 split). In the downstream, the OLT transmits downstream traffic through the splitter. Since the splitter passively distributes the optical signal from the OLT to all ONUs/ONTs on the same splitter, each ONU/ONT can physically receive the downstream transmissions destined for other ONUs/ONTs. At the transmission layer, downstream bytes are carried in GPON Transmission Convergence (GTC) frames sent periodically(per-125 μs), whose payload area carries multiple GEM frames(described next) [2], [1].

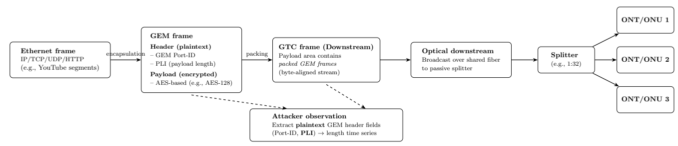


Fig. 1. Encapsulation pipeline in GPON downstream: upper-layer packets (e.g., Ethernet/IP) are encapsulated into GEM frames, packed into GTC frames, and broadcast over the shared optical line. While GEM payloads are AES encrypted (e.g., AES-128), GEM header fields such as Port-ID and PLI remain plaintext.

GEM Framing Upper layer traffic (e.g., Ethernet frames carrying TCP/IP packets) are carried in the GEM Payload Field transmitted via GPON Encapsulation Method (GEM) Frames. Downstream GEM payloads are often AES encrypted for confidentiality (depending on ISP configuration [1]). However, the GEM Port-ID (used to demultiplex services/flows) and the Payload Length Indicator (PLI), which encodes the payload size, are transmitted in plaintext. Thus, an attacker can extract a victim's encrypted payload lengths from consecutive PLI values (optionally grouped by Port-ID), even without decrypting the payload contents.

Video Identification HTTP Adaptive Streaming (HAS) is the popular video streaming model – which is generally used by video streaming platforms (e.g., Youtube) –delivers video by letting the client periodically request short media segments at a bitrate chosen by an adaptive bitrate (ABR) algorithm. This request–response process forms characteristic bursty download patterns and bitrate-dependent segment sizes, which can be exploited for traffic inference. [4]

III. Threat Model

We consider an attacker on the same optical splitter as a victim. And we assume an attacker can isolate a victim-specific GEM stream (e.g., obtain the victim's Port-ID(s)).

- **1. Premise** The attacker has an optical capture device (e.g., FPGA with optical TAP, modified ONU that can receive raw GTC frames) connected to splitter and passively records downstream from OLT.
- * The attacker does not need to register their ONU/ONT on OLT; so even if they use receive-only optical tap, they can entirely record downstream records without transmitting.
- **2. Collect GEMs** Then, attacker extracts GTC frames from downstream traffic and parses GEM frames.
- **3. Convert GEM PLI to dataset** Then, the attacker filters GEM frames for a victim's Port-ID and records PLI values with timestamps.
- **4. Dataset format** The dataset must include start & end time and duration and PLI and sequence information.
- **5. Classification via database matching** The attacker matches the captured PLI time series to a pre-collected database of PLI traces (built from known videos) and selects the closest match to infer the victim's video.

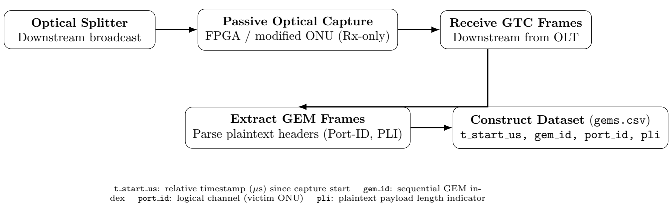


Fig. 2. Attacker-side observation pipeline: passive optical capture under the same splitter enables reception of downstream GTC frames, extraction of GEM frames, and construction of a PLI-based dataset (`gems.csv`).

IV. Experiment and Preliminary Results

To demonstrate the threat model in our experimental environment, we developed a software GPON downstream simulator that reproduces the process of receiving GTC frames, extracting GEM frames, and exporting plaintext GEM header fields into the dataset.

First, we played each of 10 videos (three runs each) on Ubuntu 24.04 under a fixed playback quality and resolution setting(720p). To minimize interference, we restricted other background traffic (e.g., VNC, SSH) while playing video. For each run, we captured packet data (PCAP) for 180 seconds. Then, we replayed each PCAP file in our simulator to record the resulting PLI values and timestamps as CSV traces. These traces approximate the payload length observations available to a real same-splitter adversary and enable systematic evaluation of PLI based inference.

After producing the dataset, we visualized cumulative PLI–Time Plot based on our dataset(Publicly Available on our github repository) [3]. We observed a clear correlation between each video and its cumulative PLI over time graph; traces from the same video show highly consistent slopes across runs.

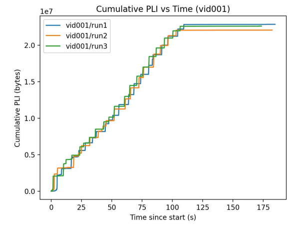
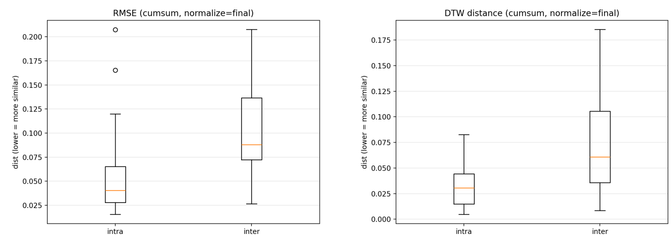


Fig. 3. PLI - Time graph for one of 10 videos showing that highly identical slopes between three runs. PLI - Time Graph of others is publicly available on our github repository [3].



(a) RMSE (lower = more similar)

(b) DTW distance (lower = more similar)

Fig. 4. Intra-video vs. inter-video similarity using normalized cumulative PLI traces. Same-video pairs produce smaller distances than different-video pairs, supporting reliable video matching from plaintext PLI sequences.

We further quantify the similarity between traces using distance-based metrics on the normalized cumulative PLI time series. Same-video pairs yield smaller distances than different-video pairs (median RMSE: 0.040 vs. 0.088; median DTW: 0.031 vs. 0.061).

V. Conclusion And Discussion

Despite GEM Payload Encryption, GPON Network is still vulnerable to inferring streaming activity because PLI in GEM headers can be exploited to infer a victim's video activity(e.g., Youtube). Future work includes validating our observations on a real GPON testbed. All code and experimental dataset is publicly available on our github repository. [3] **Mitigations** Potential mitigations include padding/obfuscating payload lengths, traffic shaping. **Future Work In Real World.** Future Work will be conducted in experimental testbed with Real OLT, ONU and Optical Splitter. We will use FPGA to collect GEM frames. **Extent to XGS-PON Networks.** We believe that XGS-PON Networks are subject to analogous privacy risks.

References

[1] ITU-T Recommendation G.984.3, *Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification*.
[2] Cisco, "Understand GPON Technology," Troubleshooting TechNotes. (Document ID: 216230)
[3] Github Repository - https://github.com/hjaka08/GPON
[4] S. Bae, M. Son, D. Kim, C. J. Park, J. Lee, S. Son, and Y. Kim, "Watching the Watchers: Practical Video Identification Attack in LTE Networks," in *31st USENIX Security Symposium (USENIX Security '22)*, 2022.
[5] X. Zhang, G. Xiong, Z. Li, C. Yang, X. Lin, G. Gou, and B. Fang, "Traffic spills the beans: A robust video identification attack against YouTube," *Computers & Security*, vol. 137, 2024, Art. 103623.
[6] N. Hu, H. Wu, H. Zhao, S. Ni, and G. Cheng, "Breaking Through the Diversity: Encrypted Video Identification Attack Based on QUIC Features," in *ESORICS 2024*, 2024.

POSTER: GEM-PLI SIDE CHANNEL IN GPON INFERRING ENCRYPTED STREAMING TRAFFIC

HyunJun An

Heungdeok High School

hjahn@anhyunjun.com

INTRODUCTION & MOTIVATION

1.As fiber internet access has become dominant in Network Infrastructures
passive optical network (PON) technologies-accounted for 54.2% of Fiber To The Home (FTTH) market in 2024
it is important to ensure traffic confidentiality.
2.GPON supports AES-based downstream traffic encryption on GEM Layer.
3.However, some fields of GEM Header remain plaintext
4.recent studies have demonstrated that such metadata like PLI can be used to infer victim's activity.
5.we researched whether the plaintext PLI field in GPON GEM header can be used to leak victim's privacy on the same optical splitter

BACKGROUND

GPON Topology and GTC Framing GPON Networks consist of an Optical Line Terminal (OLT) at the ISP and multiple Optical Network Terminals/Units(ONTs/ONUs) (typically called "modem") deployed at subscribers. A single optical tree can serve up to 32 subscribers (e.g., 1:32 split). In the downstream, the OLT transmits downstream traffic through the splitter. Since the splitter passively distributes the optical signal from the OLT to all ONUs/ONTs on the same splitter, each ONU/ONT can physically receive the downstream transmissions destined for other ONUs/ONTs. At the transmission layer, downstream bytes are carried in GPON Transmission Convergence (GTC) frames sent periodically(per-125 μs), whose payload area carries multiple GEM frames(described next).

METHODOLOGY

• 1. Premise The attacker has an optical capture device (e.g., FPGA with optical TAP, modified ONU that can receive raw GTC frames) connected to splitter and passively records downstream from OLT. • * The attacker does not need to register their ONU/ONT on OLT; so even if they use receive-only optical tap, they can entirely record downstream records without transmitting. • 2. Collect GEMs Then, attacker extracts GTC frames from downstream traffic and parses GEM frames. • 3. Convert GEM PLI to dataset Then, the attacker filters GEM frames for a victim's Port-ID and records PLI values with timestamps. • 4. Dataset format The dataset must include start & end time and duration and PLI and sequence information. • 5. Classification via database matching The attacker matches the captured PLI time series to a pre-collected database of PLI traces (built from known videos) and selects the closest match to infer the victim's video.

EXPERIMENT

we developed a software GPON downstream simulator that reproduces the process of receiving GTC frames, extracting GEM frames, and exporting plaintext GEM header fields into datasets. At first we played each of 10 videos three times in our linux based system to produce packet capture datas (PCAPs) where other streaming traffic(e.g., VNC) is restricted to minimize interference. then we simulated that PCAPs packet capture file to records the resulting PLI values and timestamps into CSV traces

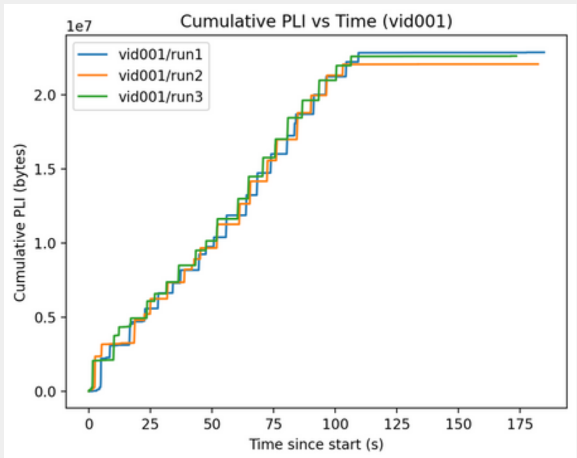


fig. PLI - Time Grapgh for one of 10 videos showing that greatly identical slopes between three runs, PLI - Time Grapgh of others is publicly available on our github repository

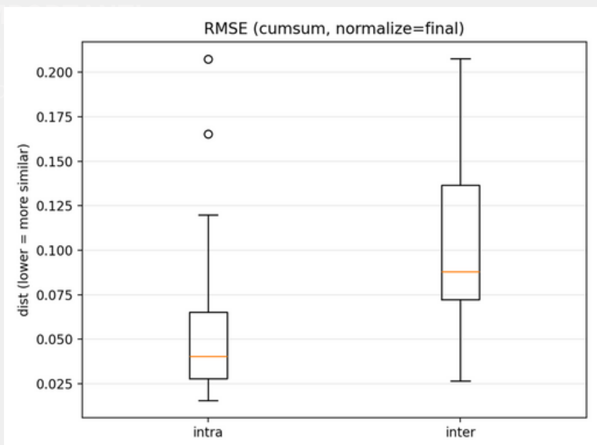


fig.RMSE of Cumulative PLI
left: Same Video
Right: Different Video

CONCLUSION

GPON Network may be vulnerable to streaming inferring attack.
Payload encryption alone does not ensure confidentiality of traffic.
RMSE: 0.040 (Same Video) / 0.088 (Different Video)

FUTURE WORK

Future Work will be conducted in experimental testbed with Real OLT and Optical Splitter. We will use FPGA to collect GEM frames.
when adopted CNN, it is possible to distinguish video Ids using captured PLI values.

DATASET & CODES

Our datasets and codes are publicly available on our github
https://github.com/hjaka08/GPON

ACKNOWLEGEMENTS

I'm a Korean High School Student, please excuse imperfect wording.
Thank you!

REFERENCES

[1] ITU-T Recommendation G.984.3, Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification.
[2] S. Bae KAIST, "Watching the Watchers: Practical Video Identification Attacks in LTE Networks," in USENIX Security, 2022.
[3] Cisco, "Understand GPON Technology," Troubleshooting TechNotes.
[4] github repository - https://github.com/hjaka08/GPON
[5] Fiber-to-the-Home (FTTH) Market Size & Share Analysis - Growth Trends and Forecast (2025 - 2030)
[6] Traffic spills the beans- A robust video identification attack against YouTube - Elsevier Computers & Security (2024)
[7] Breaking Through the Diversity: Encrypted Video Identification Attack Based on QUIC Features - ESORICS 2024